Summary in Graph

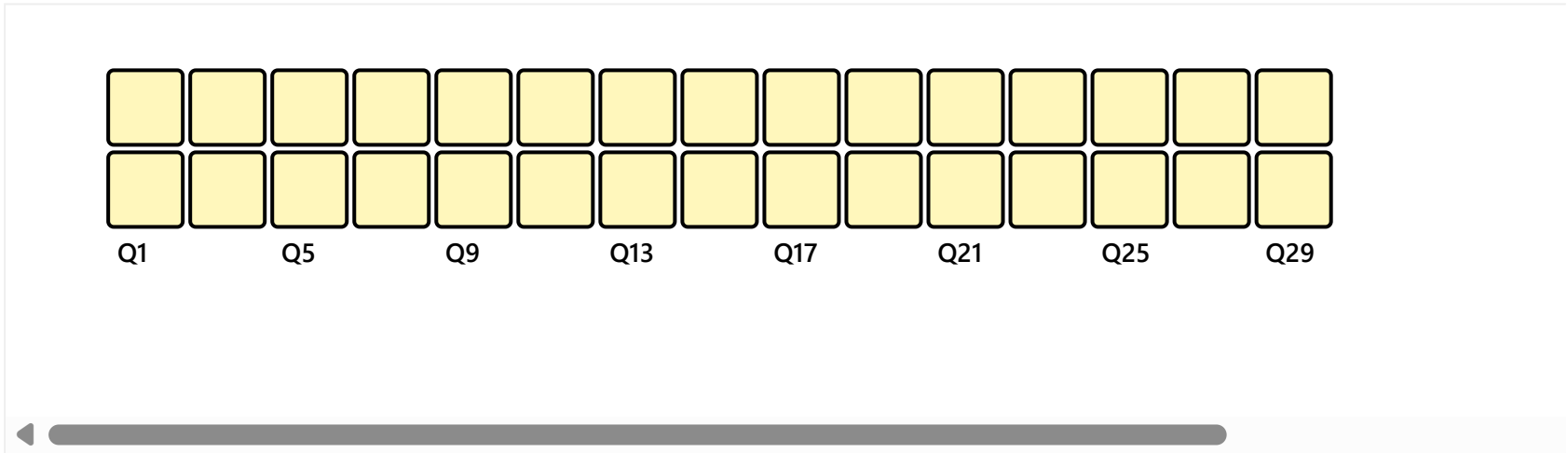
Exam Summary (GO Classes CS Test Series 2026 | Theory of Computation | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (15 + 15)
Total Marks:	45 (15 + 30)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (26) out of 256 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Below are presented two formal languages over the alphabet $\{a, b, c\}$:

$$\mathrm{L}_1 = \{a^n b^n \mid n > 1\}$$

$$\mathrm{L}_2 = \{b^n a^p c^q \mid n \geq 0, p \geq 0, q \geq 0\}$$

Consider the following statements:

I. L_1 is a regular language.

II. L_2 is a regular language.

III. L_2 is a context-free language.

Which statements are correct?

A. Only I.

B. Only II.

C. Only I and III.

D. Only II and III.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following formalisms:

I. Finite automata.

II. Finite automata with one stack.

III. Finite automata with two stacks.

Which ones contain ONLY the formalisms in which the non-deterministic variant recognizes the same set of languages as the respective deterministic version?

A. Only I.

B. Only II.

C. Only I and III.

D. I, II, and III.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Select the **INCORRECT** option.

A. The union of two recursive languages is a recursive language.

B. Let L be a recursively enumerable language. If the complement of L is recursively enumerable, then L is a recursive language.

C. The intersection of two context free languages is a context free language.

D. The union of two context free languages is a context free language.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

If A and B are languages with B regular and $\mathrm{A} \subseteq \mathrm{B}$, then

A. A must be regular.

B. A must be nonregular.

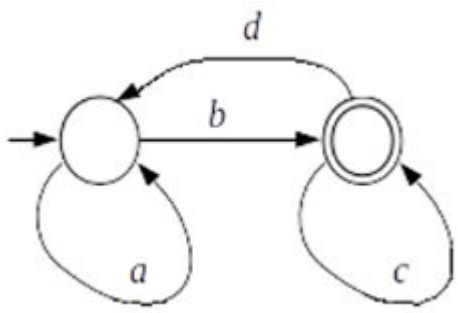
C. A must be a context-free language.

D. none of the above.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Which regular expression does the labeled graph $\mathrm{G}(\mathrm{r})$ shown below represent?



A. $r = ab^*(da^* + cb)^*$

B. $r = a^*b\,(d^* + cb)$

C. $r = (bb + d)^*(aa + c)^*$

D. $r = a^*b(c + da^*b)^*$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The class of context-free languages satisfies which of the following:

A. it is closed under intersection.

B. it is closed under complementation.

C. it is closed under concatenation.

D. it contains every possible language.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Considering the languages

$L = \{0^n 1^n 2^i \mid n \geq 0 \text{ and } i \geq 0\}$ and

$M = \{0^i 1^n 2^n \mid n \geq 0 \text{ and } i \geq 0\}$,

It can be stated that

A. the language $L \cup M$ can be generated by a context free grammar.

B. the language M can be generated by a regular grammar
.

C. The language L can be accepted by a deterministic finite automaton.

D. The language $L \cap M$ belongs to the class of context free languages.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Suppose that L_1 and L_2 are infinite regular languages. Then

A. $L_1 \cup L_2$ must be context-free.

B. $L_1 \cup L_2$ must be non-context-free.

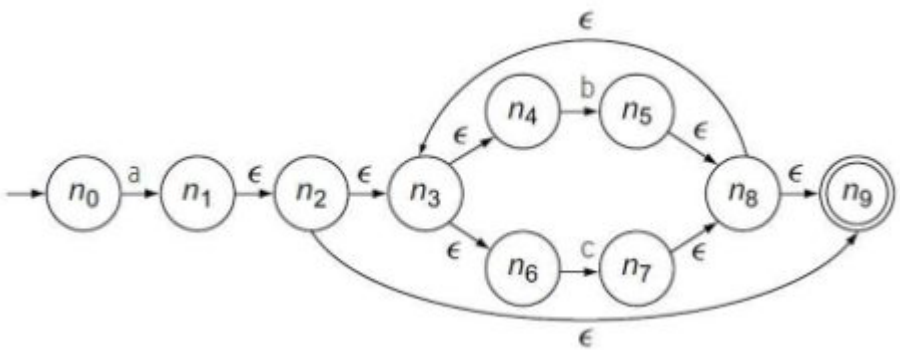
C. $L_1 \cup L_2$ can be context-free and also it can be non-context-free.

D. $L_1 \cup L_2$ must be nonregular.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following automaton.



Select the option that presents the regular expression that generates the same language recognized by the automaton.

A. $(ab)c^*$

B. $(a|b)c^*$

C. $a(b|c)^*$

D. $a(bc)^*$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Select the option that correctly presents a regular expression that denotes all strings of a's and b's that have at least two consecutive b's.

A. $(a^* + bb)(a + ba)^*(a + b)^*$

B. $(a + ba)^*bb(ba + a)^*$

C. $(a + bb)^*(bb + a)^*$

D. $(a + ba)^*bb(a + b)^*$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following context free grammar:

$$S \rightarrow aS \mid Sb \mid c$$

Which regular expression generates the same language as the grammar defined above?

A. a^*cb^*

B. a^+b^+c

C. a^+cb^+

D. ca^*b^*

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Let the languages be

$L_1 = \{a^i b^n c^m \mid i, n, m \geq 0\}$ and

$L_2 = \{a^n b^m c^i a^k \mid i, n, k, m \geq 0$, with $i = m$ or $n = m\}$.

Based on this information, it is correct to state that:

A. $L_1 \cap L_2$ is accepted by a non deterministic finite automaton.

B. $L_1 \cdot L_2$, that is, the concatenation of the languages L_1 and L_2 is not context free.

C. $L_1 \cup L_2$ is accepted by a finite automaton with at least 6 states.

D. $L_1 \cap L_2$ has a context free generating grammar.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

What is the grammar language with the following production rules

$$S \rightarrow ASb \mid c$$
$$A \rightarrow a$$

A. $\{a^n cb \mid n \in \mathbb{N}\}$

B. $\{acb^n \mid n \in \mathbb{N}\}$

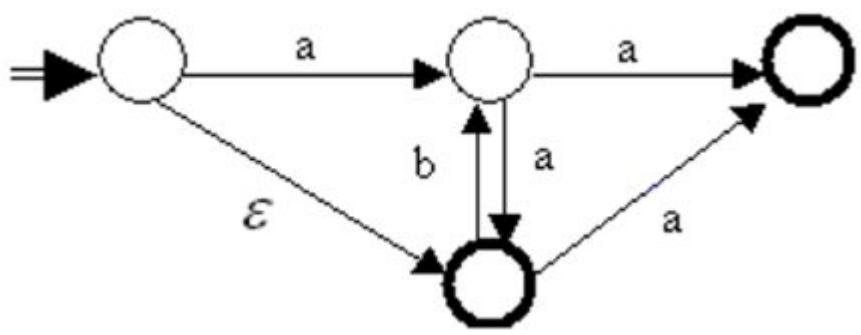
C. $\{a^n c^n b \mid n \in \mathbb{N}\}$

D. $\{a^n cb^n \mid n \in \mathbb{N}\}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the finite automaton shown in the figure below (the bold circles represent terminal states).



In this regard, mark the **FALSE** statement.

A. The word aaa is recognized by the automaton.

B. The word $ababa$ is not recognized by the automaton.

C. The empty word is recognized by the automaton.

D. The word aba is recognized by the automaton.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Mark the **INCORRECT** statement.

A. There exists a Turing machine U that simulates any other Turing machine M on any input to M.

B. Church's Thesis states that the informal concept of effective procedure is captured by the formal concept of a Turing machine.

C. A language is recursively enumerable if, and only if, it is accepted by some Turing machine.

D. There exists a Turing machine T such that, given any Turing machine M and any input w for M , T determines, in a finite number of steps, whether M halts for the input w or not.

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

The language $A = \{b^i a^j \mid i \geq 0, j \geq 0, \text{ and } j = 2i\}$ satisfies which of the following?

A. A has regular expression $b^*(aa)^*$.

B. A has regular expression $(baa)^*$.

C. A has CFG $G = (V, \Sigma, R, S)$, with $V = \{S\}, \Sigma = \{a, b\}, R = \{S \rightarrow bSaa\}$, and starting variable S.

D. none of the above.

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the following grammar G, where S is the start symbol:

$\mathrm{S} \rightarrow \mathrm{AcB}$

$\mathrm{A} \rightarrow \mathrm{cA} \mid \mathrm{aB}$

$\mathrm{B} \rightarrow \mathrm{cB} \mid \mathrm{aA}$

$\mathrm{A} \rightarrow \varepsilon$

Select the option(s) that present the word that belongs to the language generated by grammar G.

A. aaca

B. aaaca

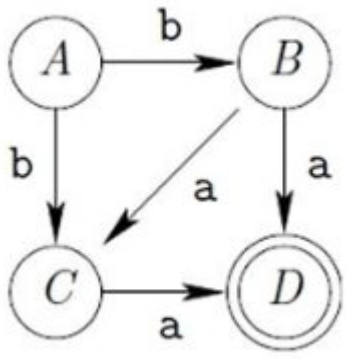
C. ccac

D. aaa

Your Answer: | **Correct Answer: A** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following non-deterministic finite automaton, where A is the initial state and D is the only accepting state.



Which deterministic finite automaton with d as its state transition function accepts the same language?

A. Initial state A, accepting states C and D
$d(\mathrm{A}, \mathrm{b}) = \mathrm{B}$
$d(\mathrm{B}, \mathrm{a}) = \mathrm{C}$
$d(\mathrm{C}, \mathrm{a}) = \mathrm{D}$

B. Initial state A, accepting state C
$d(\mathrm{A}, \mathrm{b}) = \mathrm{B}$
$d(\mathrm{B}, \mathrm{a}) = \mathrm{C}$
$d(\mathrm{C}, \mathrm{a}) = \mathrm{C}$

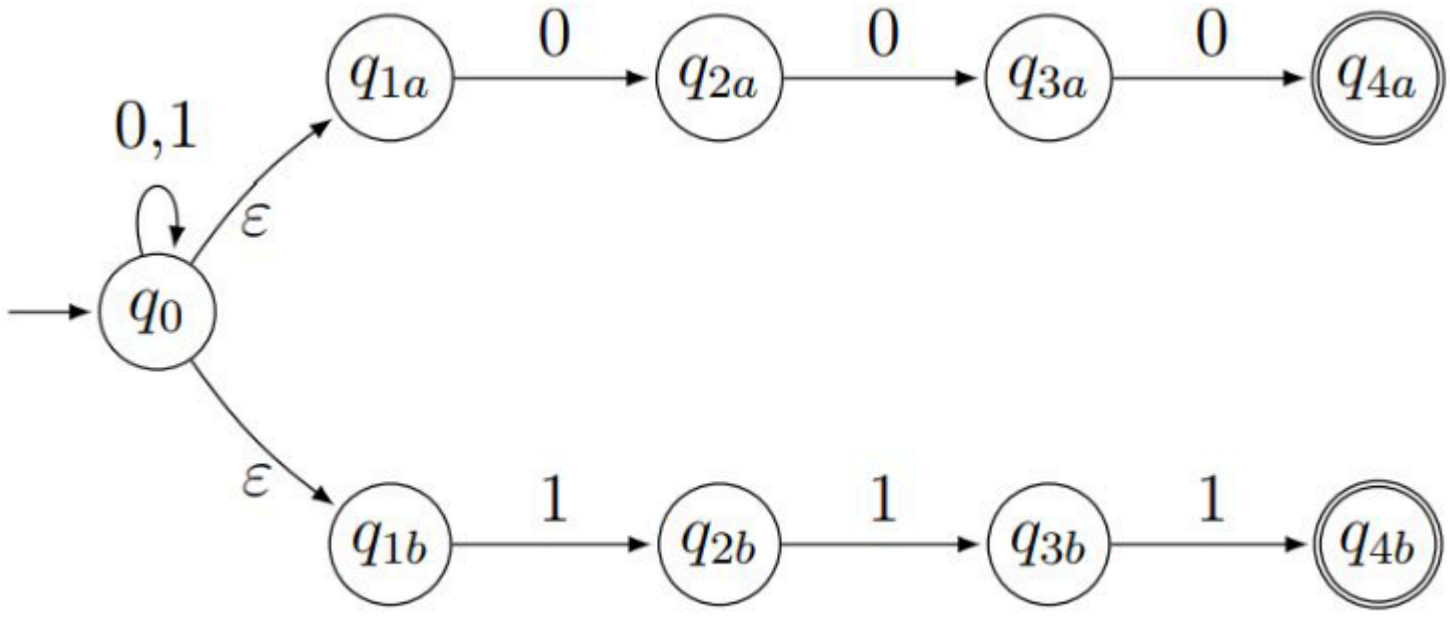
C. Initial state A, acceptance state D
$d(\mathrm{A}, \mathrm{b}) = \mathrm{B}$
$d(\mathrm{B}, \mathrm{a}) = \mathrm{D}$
$d(\mathrm{B}, \mathrm{b}) = \mathrm{C}$
$d(\mathrm{C}, \mathrm{a}) = \mathrm{D}$

D. It is impossible to convert this non-deterministic finite automaton into a deterministic finite automaton.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following non-deterministic automaton.



Let p be the probability that a uniformly random string from $\{0,1\}^{10}$ is accepted by the above automaton. Which of the following is true about p ?

A. $0 < p \leq 0.05$

B. $0.05 < p \leq 0.10$

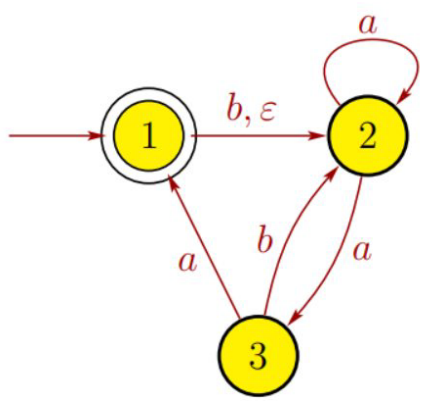
C. $0.10 < p \leq 0.15$

D. $p > 0.20$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Which choice below is correct for the following NFA N?



A. $ab \in L(N)$.

B. $\varepsilon \in L(N)$.

C. $\varepsilon \notin L(N)$.

D. none of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following property about a formal language L :

"There exists a number $p \geq 0$, such that for any word $w \in \mathrm{L}, |w| \geq p$, there exist words x, y, and z, with $\mathrm{y} \neq \varepsilon$ and $|xy| \leq p$, for any integer $\mathrm{i} \geq 0$, the word $xy^iz \in \mathrm{L}$."

Based on the statement and your knowledge of the subject, mark T (true) or F (false) for the following statements.

( ) If L is accepted by a non-deterministic finite automata , then L satisfies the above property.

( ) If L is not regular, L doesn't satisfy the above property.

( ) The language $\{a^nb^nc^n \mid n \geq 0\}$ does not satisfy the above property.

( ) If L doesn't satisfy the above property, L is not regular,

Select the option that contains, from top to bottom, the correct sequence.

A. $\mathrm{T}, \mathrm{T}, \mathrm{T}, \mathrm{T}$.

B. $\mathrm{T}, \mathrm{F}, \mathrm{T}, \mathrm{T}$.

C. $\mathrm{T}, \mathrm{F}, \mathrm{F}, \mathrm{T}$.

D. $\mathrm{T}, \mathrm{F}, \mathrm{T}, \mathrm{F}$.

Your Answer: | **Correct Answer: B** | Not Attempted | Time taken: 00min 03sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Regarding grammars and languages, consider the following statements

I. A grammar in the Chomsky Normal Form can be ambiguous.

II. There is some context free language for which no ambiguous context free grammar exists.

III. A grammar in Greibach Normal Form can be converted to Chomsky Normal Form.

IV. The algorithm for converting a Context Free Grammar to a Chomsky Normal Form Grammar can be directly applied to a grammar that is not λ free.

Select the correct option.

A. Only statements I and II are correct.

B. Only statements I and IV are correct.

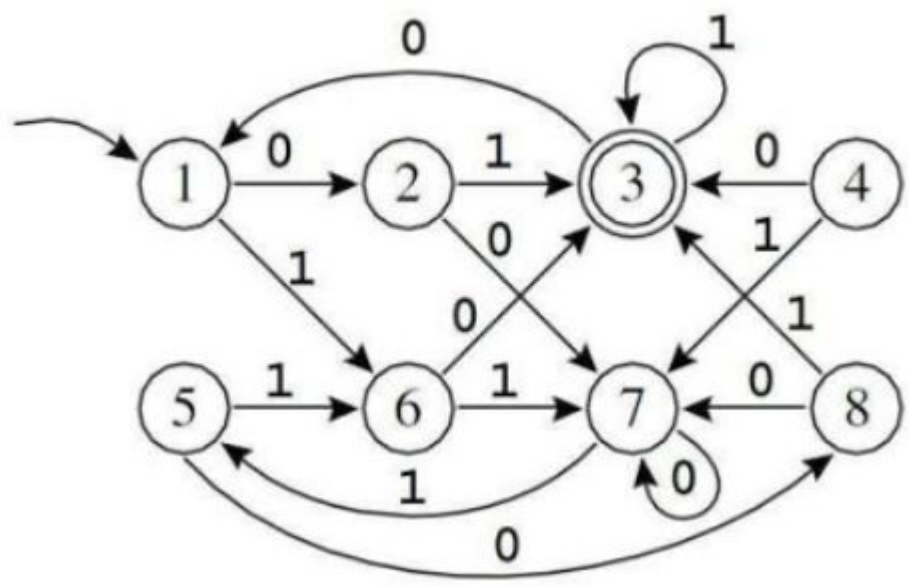
C. Only statements III and IV are correct.

D. Only statements I, II, and III are correct.

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following automaton.



Regarding this automaton, consider the following statements.

I. States 3 and 7 are equivalent.

II. States 4 and 6 are equivalent.

III. States 1 and 5 are equivalent.

IV. States 2 and 8 are equivalent.

Select the correct option.

A. Only statements I and IV are correct.

B. Only statements III and IV are correct.

C. Only statements I, II, and III are correct.

D. Only statements II, III, and IV are correct.

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

If a finite number of strings is added to a nonregular language A , then the resulting language B satisfies which of the following?

A. B must be a regular language.

B. B must be a nonregular language.

C. B must be a non-context-free language.

D. B must have a context-free grammar.

Your Answer: | **Correct Answer: B** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Let $\Sigma = \{0, 1\}$, and let L be the language of all non-empty strings over Σ that begin and end with the same symbol. Consider the following regular expressions:

$\mathbf{R1}$: 01^*0

$\mathbf{R2}$: $0(0 \cup 1)^*0 \cup 1(0 \cup 1)^*1 \cup (0 \cup 1)$

$\mathbf{R3}$: $(0 \cup 1)(0 \cup 1)^*(0 \cup 1)$

Which of the following statements is correct?

A. $\mathrm{R1}$ generates L, but $\mathrm{R2}$ and $\mathrm{R3}$ do not.

B. $\mathrm{R2}$ generates L, but $\mathrm{R1}$ and $\mathrm{R3}$ do not.

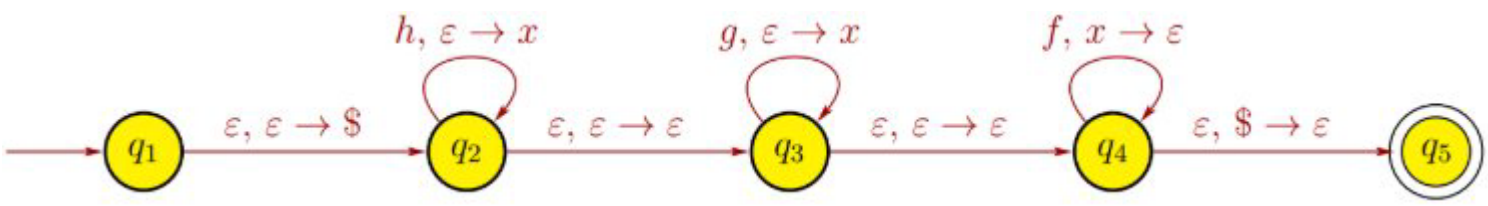
C. $\mathrm{R3}$ generates L, but $\mathrm{R1}$ and $\mathrm{R2}$ do not.

D. $\mathrm{R1}$ and $\mathrm{R2}$ generate L, but $\mathrm{R3}$ does not.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

For $\Sigma = \{f, g, h\}$, let $A \subseteq \Sigma^*$ be the language recognized by the following PDA:



Consider the following languages:

I. $L_1 = \{h^i g^j f^k \mid i, j, k \geq 0 \text{ and } i = j + k\}$

II. $L_2 = \{h^i g^j f^k \mid i, j, k \geq 0 \text{ and } i + k = j\}$

III. $L_3 = \{h^i g^j f^k \mid i, j, k \geq 0 \text{ and } i = j = k\}$

Which of the following statements is correct?

A. $A = L_1, A \neq L_2$, and $A \neq L_3$.

B. $A \neq L_1, A = L_2$, and $A \neq L_3$.

C. $A \neq L_1, A \neq L_2$, and $A = L_3$.

D. $A \neq L_1, A \neq L_2,$ and $A \neq L_3$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Analyze the following statements about automata and languages:

I. Deterministic finite automata and non deterministic finite automata accept the same set of languages.
II. Every recursive language is also a recursively enumerable language.
III. Let L be a context free language. There exists a deterministic pushdown automata that recognizes L.

Which are correct?

A. Only I.

B. Only II.

C. Only I and II.

D. Only I and III.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

The language $A = \{w \in \{0,1\}^* \,|w = w^{\mathcal{R}}, |w| \text{ is even } \}$ satisfies which of the following?

A. A has regular expression $(0^*1^*)^*(1^*0^*)^*$.

B. A has context-free grammar $G = (V, \Sigma, R, S)$, with $\Sigma = \{0,1\}, V = \{S\}$, starting variable S, and rules $R = \{S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \varepsilon\}$.

C. A has context-free grammar $G = (V, \Sigma, R, S)$, with $\Sigma = \{0,1\}, V = \{S\}$, starting variable S, and rules $R = \{S \rightarrow 0S0 \mid 1S1 \mid \varepsilon\}$.

D. all of the above.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the grammar $\mathbf{G}$ described below: set of terminals $\{\mathbf{a}, \mathbf{c}\}$, set of non-terminals $\{\mathbf{S}, \mathbf{A}\}$, initial symbol $\mathbf{S}$, and containing the following productions:

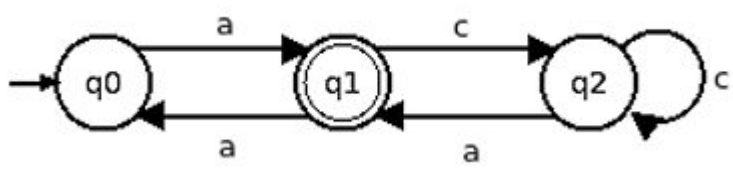
$\mathrm{S} \rightarrow \mathrm{AcS}$

$\mathrm{S} \rightarrow \mathrm{A}$

$\mathrm{A} \rightarrow \mathrm{aAa}$

$\mathrm{A} \rightarrow \mathrm{a}$

Also consider the finite automaton A over the alphabet $\{\mathbf{a}, \mathbf{c}\}$, with state set $\{\mathbf{q0}, \mathbf{q1}, \mathbf{q2}\}-$ where $\mathbf{q0}$ is the initial state and $\mathbf{q1}$ is the final state $-$ and with state transition function determined by the following graph.



Let $\mathbf{L}(\mathbf{G})$ be the language generated by grammar $\mathbf{G}$ and $\mathbf{L}(\mathbf{A})$ be the language recognized by automaton $\mathbf{A}$.

Select the correct option.

A. $\mathbf{L}(\mathbf{G})$ is regular and $\mathbf{L}(\mathbf{A})$ is a proper subset of $\mathbf{L}(\mathbf{G})$.

B. $\mathbf{L}(\mathbf{G})$ is not regular and $\mathbf{L}(\mathbf{A})$ is a proper subset of $\mathbf{L}(\mathbf{G})$.

C. $\mathbf{L}(\mathbf{A}) = \mathbf{L}(\mathbf{G})$.

D. $\mathbf{L}(\mathbf{G})$ is regular and $\mathbf{L}(\mathbf{G})$ is a proper subset of $\mathbf{L}(\mathbf{A})$.

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

If A and B are finite languages defined over the same alphabet Σ, then

A. $\bar{A} \circ B$ must be finite.

B. $\bar{A} \circ B$ must be infinite.

C. $A \cap \bar{B}$ must be context-free.

D. $\bar{A} \cup \bar{B}$ must be nonregular.

Your Answer: | **Correct Answer: C** | Not Attempted | Time taken: 00min 00sec | Discuss

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